

5.5 Fourth Test: Vacuum LiCN

For the final test, a system of 11 states will be used to simulate the propagation of a LiCN molecule in the vacuum. The only transition operators considered in this implementation are $\hat{S}_{1,0} = \sqrt{\Gamma_{0,1}} |0\rangle\langle 1|$ and $\hat{S}_{2,0} = \sqrt{\Gamma_{0,2}} |0\rangle\langle 2|$, where $\Gamma_{0,1} = 0.001$ and $\Gamma_{0,2} = 0.004$. The averages over the stochastic noise are computed by averaging over 102 paths. The initial population of the first six states are $1/6$ and the initial population of the last five states are 0.0. This system was used as a test for the quantum jump method, implemented also in WaveT. The results comparing the Euler-Maruyama, Heun and Runge-Kutta (4th order) schemes with the quantum jump method are shown in Figure 10 for a time step of $\Delta t = 5 \times 10^{-2}$.

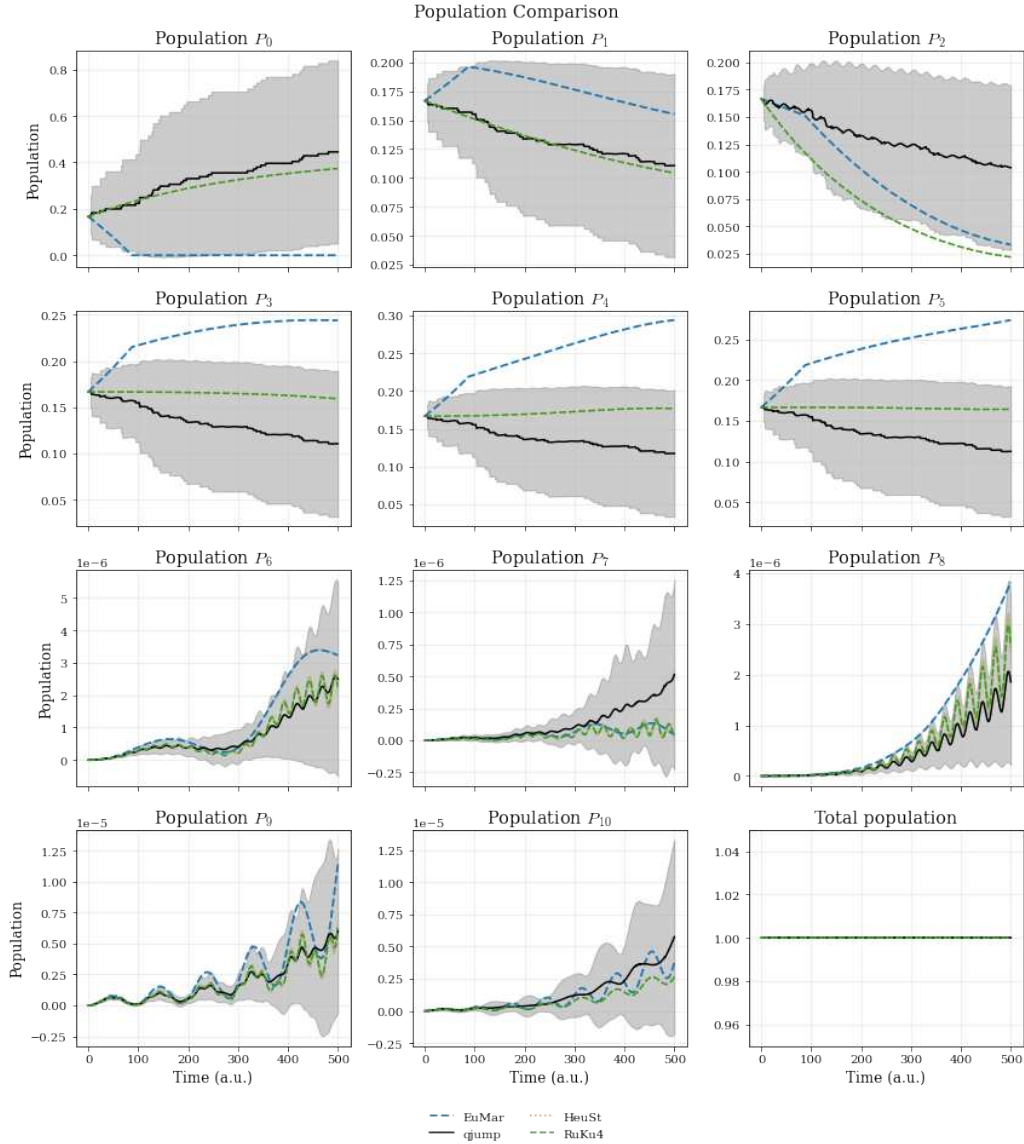


Figure 10: Results for the implementation of the stochastic Schrödinger equation (SSE) using a system of 11 states with a time step of $\Delta t = 5 \times 10^{-2}$ for all schemes considered in this section (Euler-Maruyama, Heun, Runge-Kutta (4th order) and quantum jump method). The Euler-Maruyama, Heun and Runge-Kutta (4th order) schemes are compared with the quantum jump method.

The Euler-Maruyama scheme gave the worst results among the three schemes tested. For this reason, new results were obtained with the time step decreased to $\Delta t = 5 \times 10^{-3}$ only for this method. The results are shown in Figure 11.

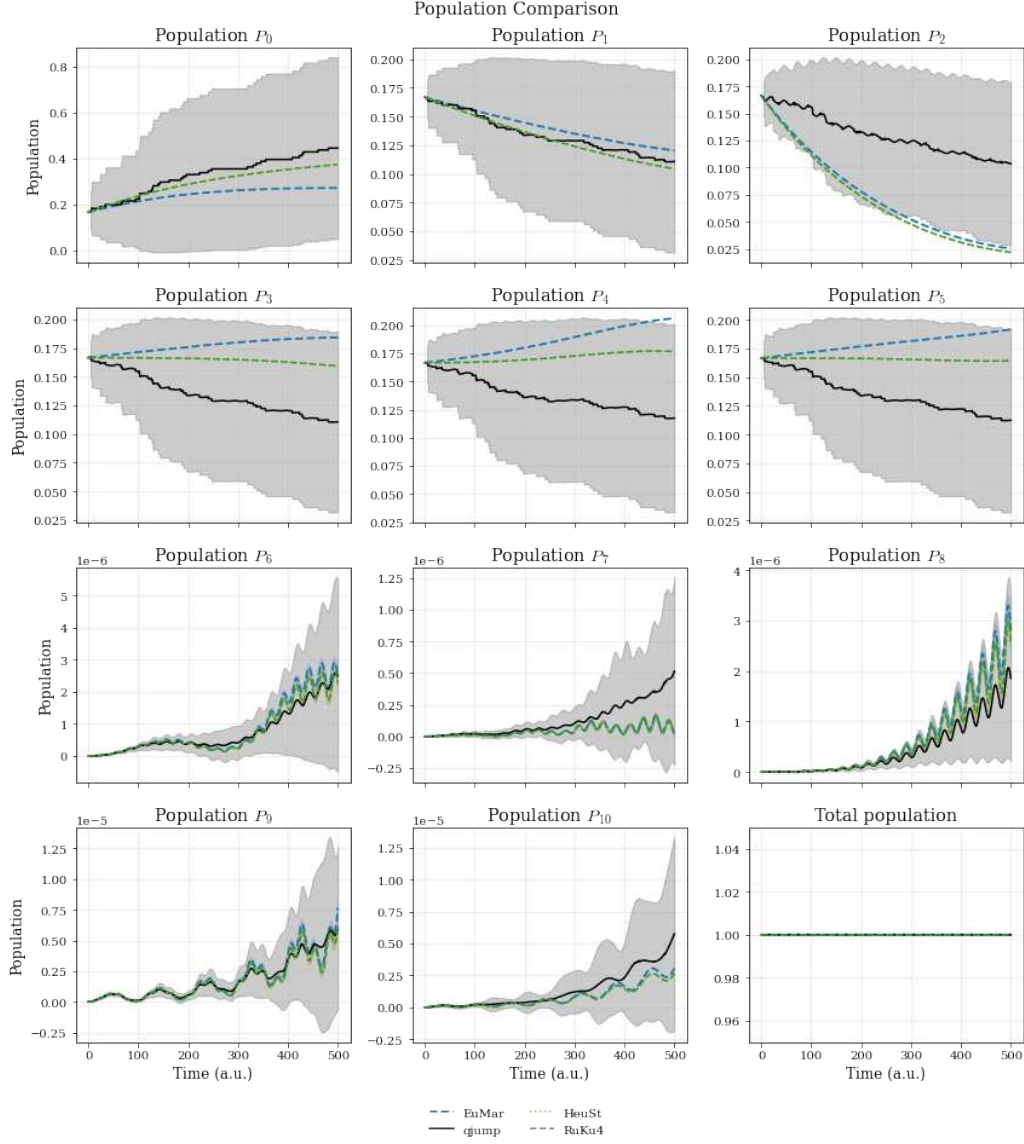


Figure 11: Results for the implementation of the stochastic Schrödinger equation (SSE) using a system of 11 states with a time step of $\Delta t = 5 \times 10^{-3}$ for the Euler-Maruyama scheme, and $\Delta t = 5 \times 10^{-2}$ for the other schemes.